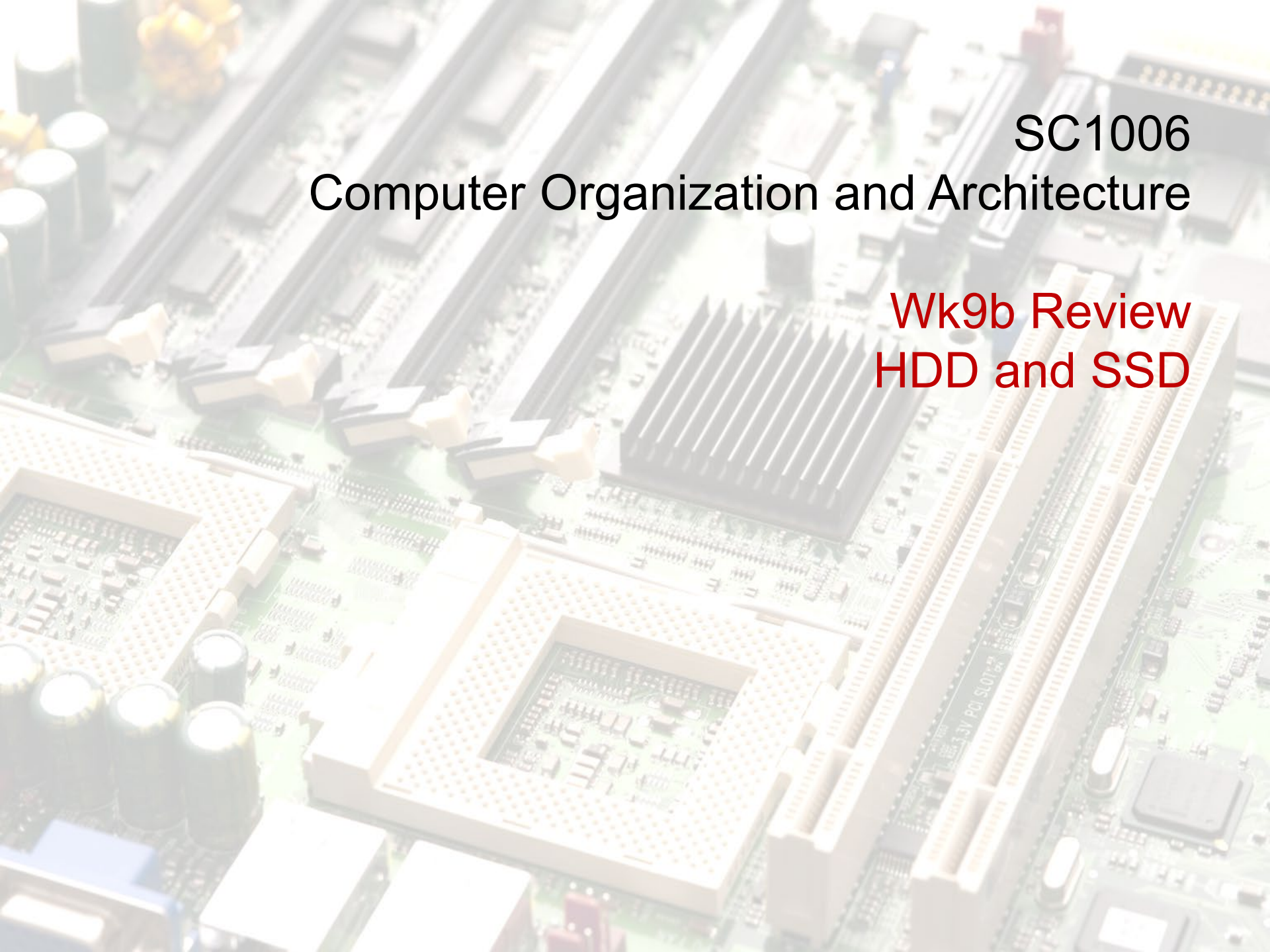




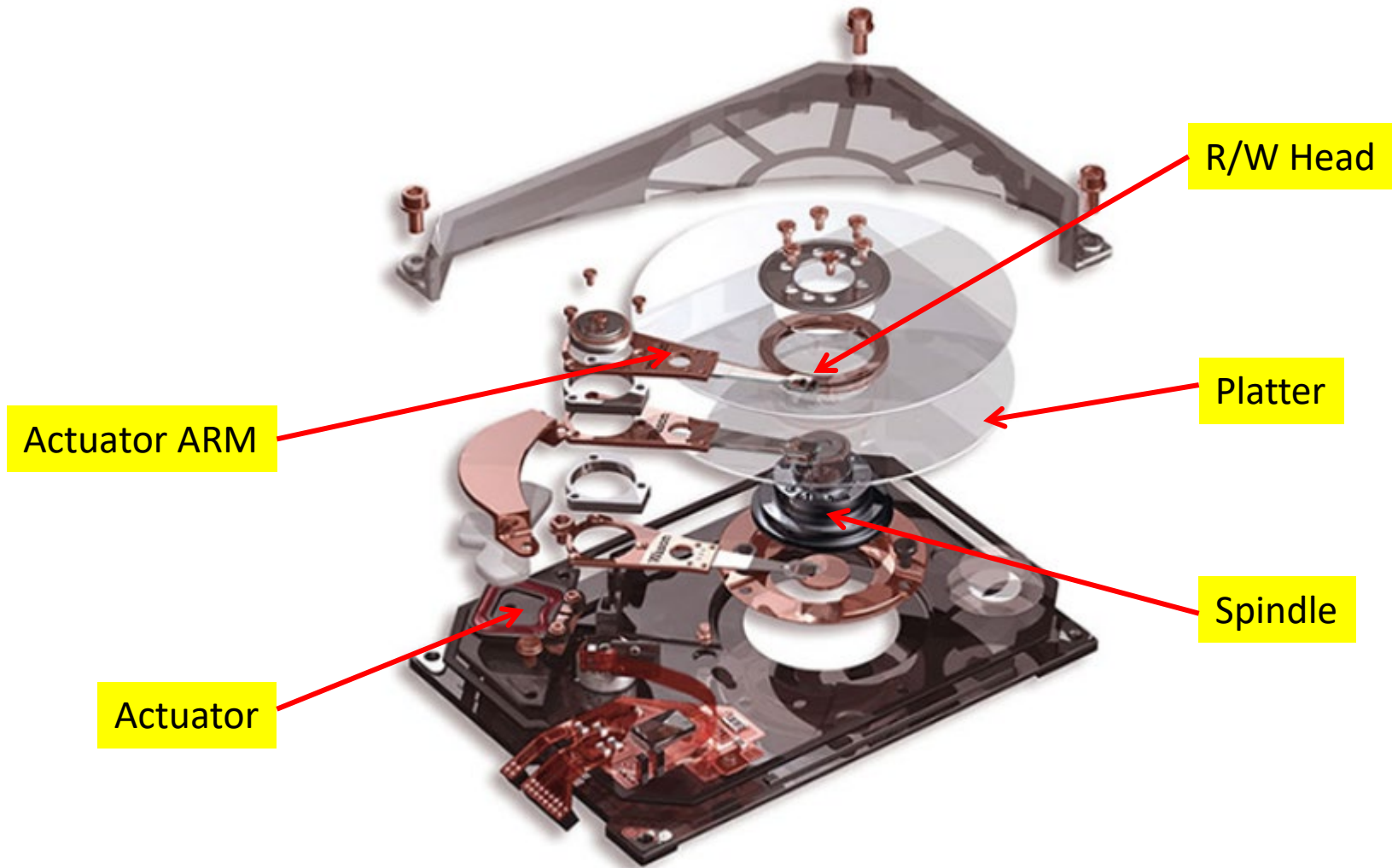
SC1006

Computer Organization and Architecture

Wk9b Review
HDD and SSD

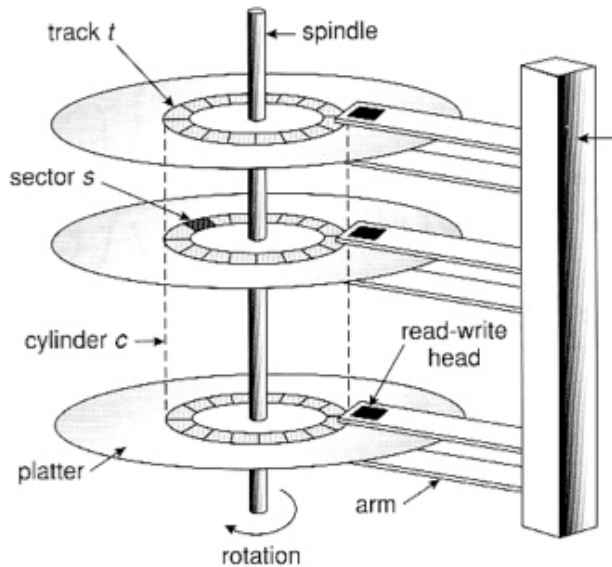
HDD AND SSD

Magnetic Hard Disk



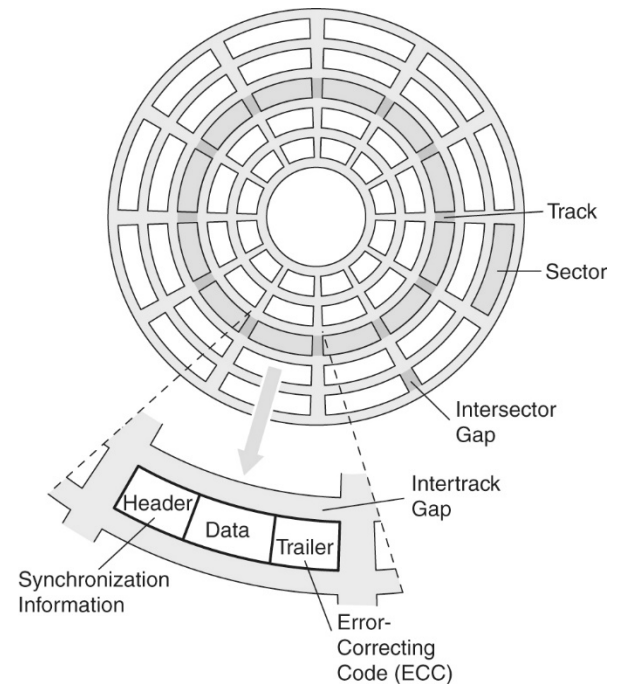
Source: Backblaze

Magnetic HDD Data Organization



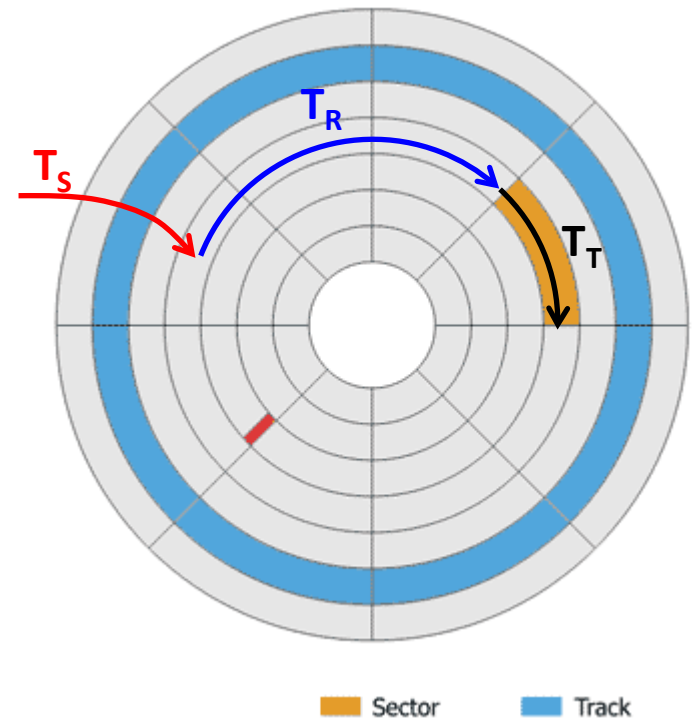
- Computers often use magnetic hard disks for large secondary storage devices.
 - One or more **platters** on a common spindle.
 - Platters are covered with thin magnetic film.
 - Platters rotate on **spindle** at constant rate.

- **Data** is stored on the **surface** of the platter in concentric rings called **tracks**.
 - Gaps between tracks to minimize interferences from adjacent tracks.
- **Tracks** are divided into **sectors**.
 - Minimum data block size is a sector.
- Stored data consists **header, data and trailer**.



HDD Transfer Rate

- Seek time (T_S)
 - Time taken for the head to move to the correct track.
- Rotational Delay (T_R)
 - Time taken for the disk to rotate until the read/write head reaches the starting position of the target sector.
- Access Time (T_A)
 - Time from request to the time the head is in position ($T_S + T_R$)
- Transfer Time (T_T)
 - Time required to transfer the required data after the head is positioned.

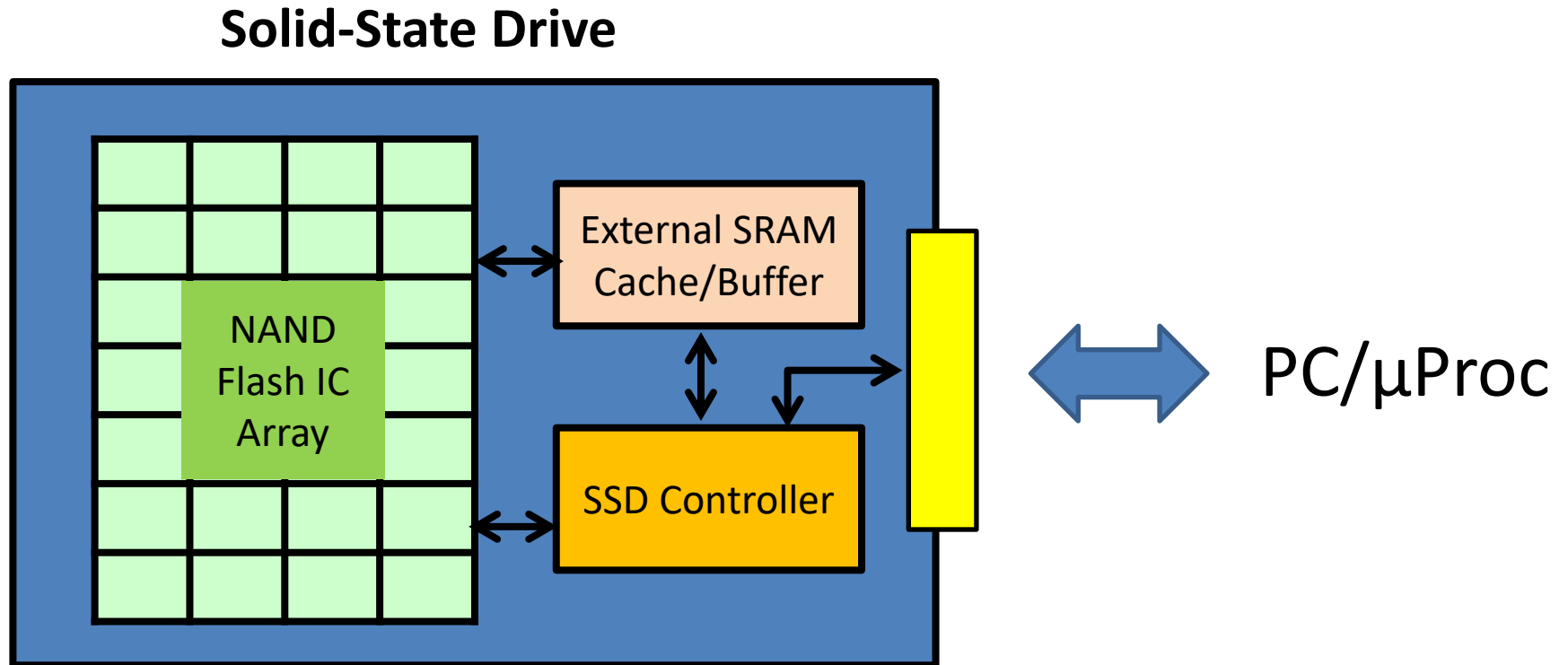


Solid State Drive (SSD)

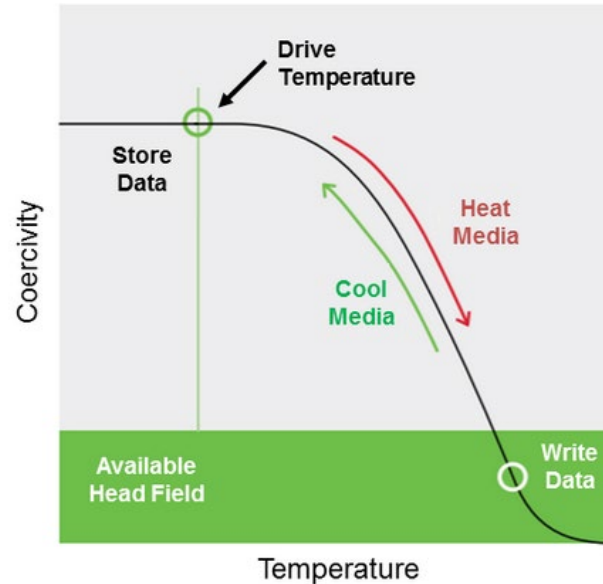
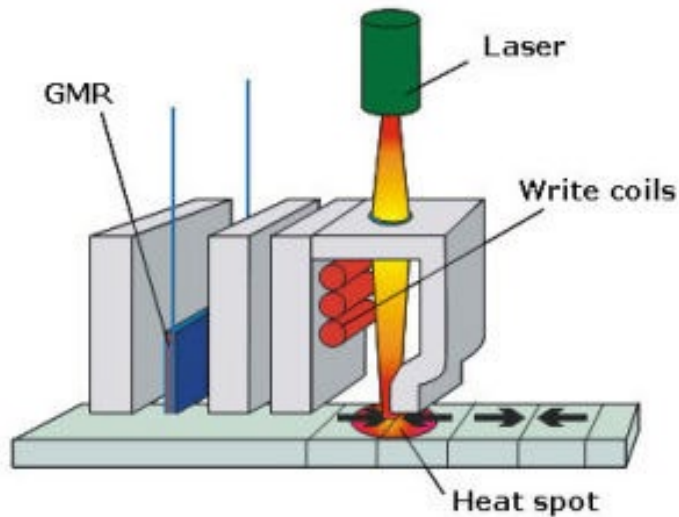


- Solid-state drives (SSD) are becoming popular
 - Memory array based on **NAND-FLASH** or NOR-FLASH.
 - Still more expensive than magnetic hard-disk.
- Flash memory has **limited program-erase cycles** (3,000 to 1,000,000).
- Various techniques used to **extend the life of SSD**
 - **Wear levelling** is a technique used to extend the life of the SSD disk by distributing data erase/write operations evenly over the entire disk.
 - **Use External RAM** as buffer to minimize the number of writes to Flash in SSD.
 - **Error Correction Code**. Ability to recover from one or more bits of error in media.
- **MTBF** (Mean Time To Failure) of recent SSD is comparable to that of HDD.

SSD System Block Diagram



HAMR (Heat Assisted Magnetic Recording)



- The **areal density** of the HDD was **stagnant** for a while and manufactures had been shipping drives with 2TByte/Platter.
- But with the **HAMR**, the areal density is expected to **increase** again.
- A **small laser** is attached to a **recording head**, designed to heat a tiny spot on the disk where the data will be written. This allows a smaller bit cell to be written as either a 0 or a 1.
- Current projections are that HAMR can achieve **5 Tbps**, enabling hard drives with capacities higher than **100 TB**.

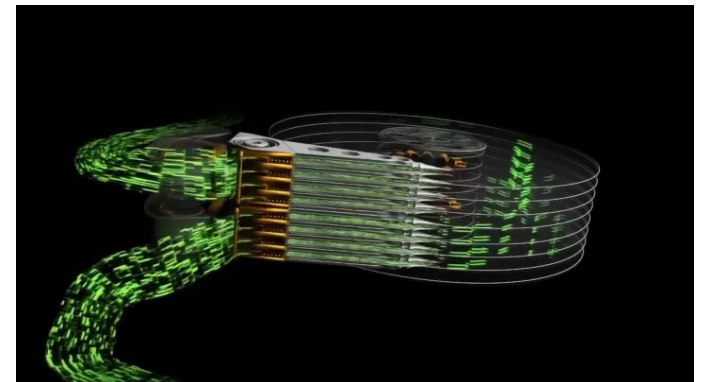
HAMR and Dual-Actuator in Action



HAMR Drive



Dual Actuator



Previous thinking questions

We have entered the digital era.
Why do we still use analog-
based/related sensors, amplifiers,
filters, converters (ADC, DAC),
etc?



New thinking questions toward understanding...

What are the relative advantages and disadvantages of SSD against HDD?

A real-world company request for project recently: to prolong new-gen SSD's life, optimize its space use, "output" its spare computing power, ...

